7CS039 Statistics for AI & Data Science

Week 3 Correlation & Regression

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1 Introduction

It frequently happens that two different "measurements" (x and y) are made for each "subject" in a study. Such data may be displayed in a "scatter diagram" (also known as a scatterplot, scattergram, or scattergraph), an example is shown in Figure 1

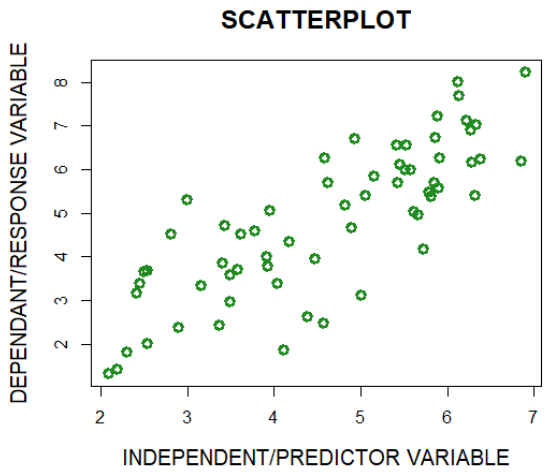

Figure 1: Scatterplot or scatter diagram

N.B. If one variable comes first in time, or one is independent and one dependent (not necessarily causally), the earlier or independent variable, usually (but far from exclusively) called x, is drawn on the horizontal axis; the later or dependent variable is (usually) called y and is drawn on the vertical axis. It is important to note that often which variable is "independent" and which dependent is quite arbitrary. For example, if we were to have data relating to the heights of 25 married couples it is not that either "height of husband" depends on "height of wife" or vice versa.

In such settings, two (related but distinct) questions arise naturally:

Q1. Is there a linear relationship between x and y? How strong is any such relationship?

Q2. Assuming there is a linear relationship, what is it?

The first question relates to "correlation" and the second question relates to "regression".

Both ideas (correlation and regression) can be extended to more than two variables - , and to non-linear relationships, i.e. to relationships other than straight lines, this is known as multiple and curvilinear regression/ correlation.

2 Correlation

Various measures of correlation have been developed for different kinds of data. All run from $+1$ (exact direct relationship, as x increases y also increases) to -1 (exact reverse relationship, as x increases y decreases and vice versa). A correlation coefficient of 0 indicates no linear relationship.

The most common correlation coefficient is Pearson's product-moment correlation coefficient. It is denoted by the Greek letter ρ (if we are referring to the population correlation), or by r if we are referring to a correlation calculated from a sample in order to estimate the population correlation. (This requires interval or ratio level data). The sample correlation may be calculated by:

$$r = \frac{\sum_{i=1}^{n}(X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_{i=1}^{n}(X_i - \bar{X})^2}\sqrt{\sum_{i=1}^{n}(Y_i - \bar{Y})^2}}$$

Nowadays however it is invariably calculated using a computer package, many calculators have functions to calculate the correlation coefficient also.

Example

For example, consider the following table which records the amounts of crates of beer drank in a given bar, and the temperature (in Fahrenheit), over a ten day period.

Day	1	2	3	4	5	6	7	8	9	10
Temperature	55	60	60	70	62	64	68	71	63	62
Beer Sales	40	41	43	49	44	45	49	50	44	43

Example ctd.

Let us look at a scatter plot of the variables "temperature" and "beer sales".

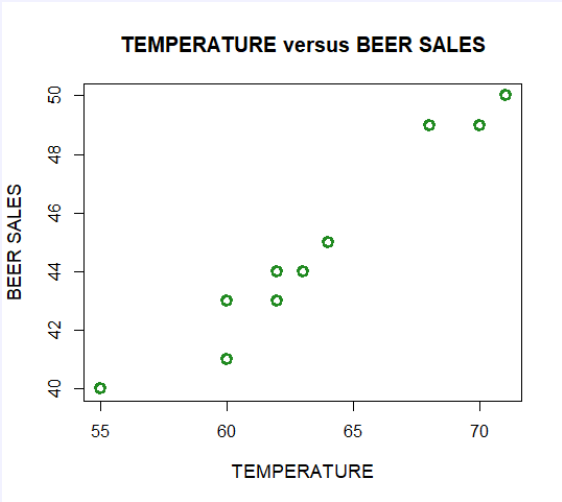

Figure 2: Strong, Positive, Linear Correlation

From the scatter plot we see that:

- Beer sales increase as temperature increases. [*Positive* correlation]
- If we drew the straight line on the scatter plot that best fitted the points this line would nearly pass through all the points. [*Strong linear* correlation]

Scatterplots for "temperature" and "coffee sales" and "temperature" and "bread sales" might be as follows:

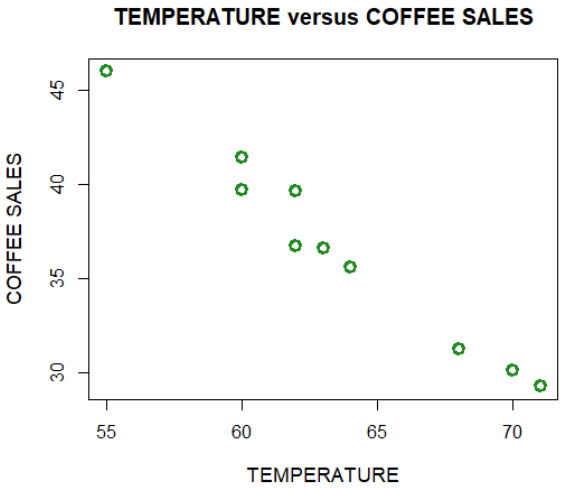

Figure 3: Strong, Negative, Linear Correlation

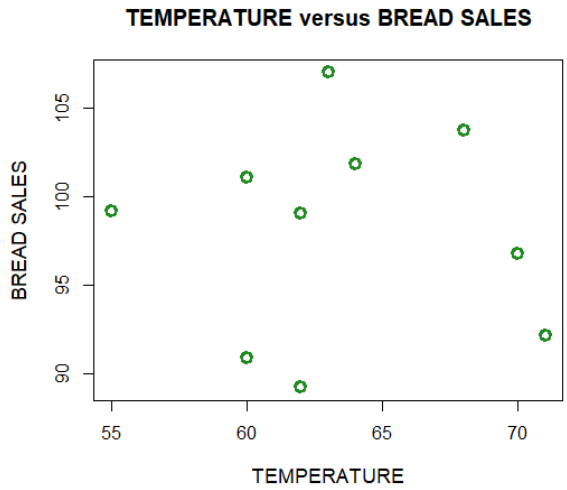

Figure 4: Little or No Linear Correlation

Note that two variables that may be weakly *linearly* correlated may be strongly correlated some other way. e.g. the variables below are related by an equation of the form $Y = aX^2 + bX + c + \varepsilon$, hence they are strongly quadratically correlated.

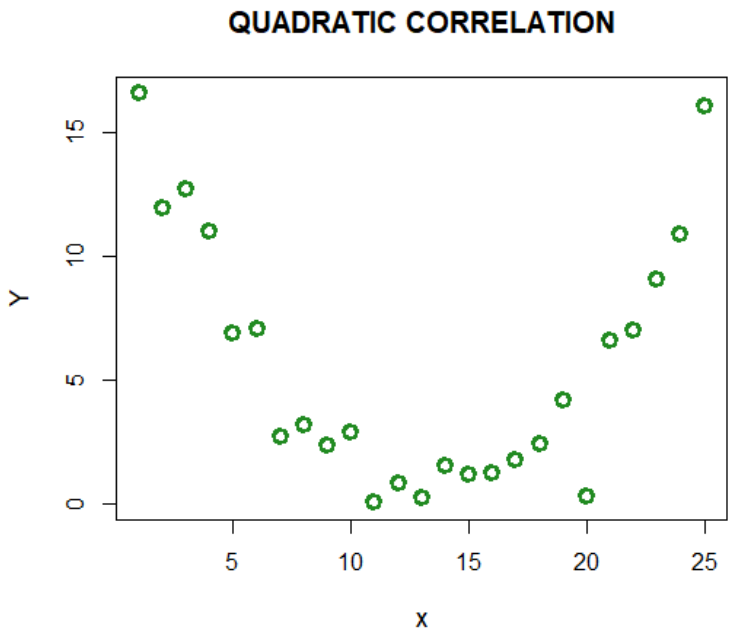

Figure 5: Strong Quadratic Correlation (but negligible linear correlation)

In layman's terms, two variables are strongly correlated if knowledge of the value of one variable enables precise prediction of the value of another, and weakly correlated if knowledge of the value of one variable enable some, but poor, prediction of the value of the other.

2.1 Correlation Does not (necessarily) Imply Causation

"Correlation does not imply causation" is a phrase used in science and statistics to emphasize that correlation between two variables does not automatically imply that one causes the other. The opposite belief, correlation proves causation, is a logical fallacy by which two events that occur together are claimed to have a cause-and-effect relationship. For example, in the UK there is a moderate negative correlation between temperature and sales, in general there is a higher volume of sales when it is colder. This is however due to cold weather tending to occur around Christmas. Similarly, until recently there was a strong negative correlation between the number of televisions in a country and the infant mortality rate in that country, but this was a consequence of both being influenced by poor economic conditions.

2.2 (Pearson's) Correlation Coefficient

We most commonly measure correlation using Pearson's correlation coefficient. Pearson's correlation coefficient should only be used between two variables that are measured on an interval scale. Other versions exist for when this is not the case, but we shall not cover them in this course.

2.2.1 Some properties of Pearson's Correlation Coefficient

(i) $-1 \leq r \leq 1$

(ii) (a) $r < 0 \longleftrightarrow$ negative correlation

(b) $r > 0 \longleftrightarrow$ positive correlation

(c) $r = 0 \longleftrightarrow$ zero correlation

(iii) (a) $r = 1 \longleftrightarrow$ perfect positive correlation

(b) $r = -1 \longleftrightarrow$ perfect negative correlation

(iv) r is the *sample* (linear) correlation coefficient, and hence it will vary according to the particular sample chosen. The *population* (linear) correlation coefficient is denoted by the Greek letter ρ, (rho). Methods exist for estimating confidence intervals for ρ.

For example, using any statistical software program we may determine that for the "beer sales example" $r = 0.974$, indicating strong, positive, linear correlation.

Example

Blood is a mixture of cells and plasma. The packed cell volume (PCV) is a measurement of the proportion of blood that is made up of cells. The value is expressed as a percentage or fraction of cells in blood. For example, a PCV of 40% means that there are 40 millilitres of cells in 100 millilitres of blood. Haemoglobin (Hb) is a protein found in red blood cells in your blood and is a good indication of your blood's ability to carry oxygen throughout your body. A natural question is whether there is a relationship between a person's age, their PCV levels and their Hb level. Table 1 contains relevant data, Table 2 describes the correlations between the variables and Figure 6 contains the scatterplots. We see that their is a moderate-strong linear correlation between PCV and Age, the other correlations being weak.

Table 1: Data from Campbell et al (1985)

Subject No.	Age	Hb (g/dl)	PCV (%)
1	20	11.1	35
2	22	10.7	45
3	25	12.4	47
4	28	14.0	50
5	28	13.1	31
6	31	10.5	30
7	35	12.5	33
8	38	13.5	35
9	40	13.9	40
10	45	15.1	45

Table 2: Correlations

Age with Hb	$r = 0.7242$
Age with PCV	$r = -0.0494$
Hb with PCV	$r = 0.3654$

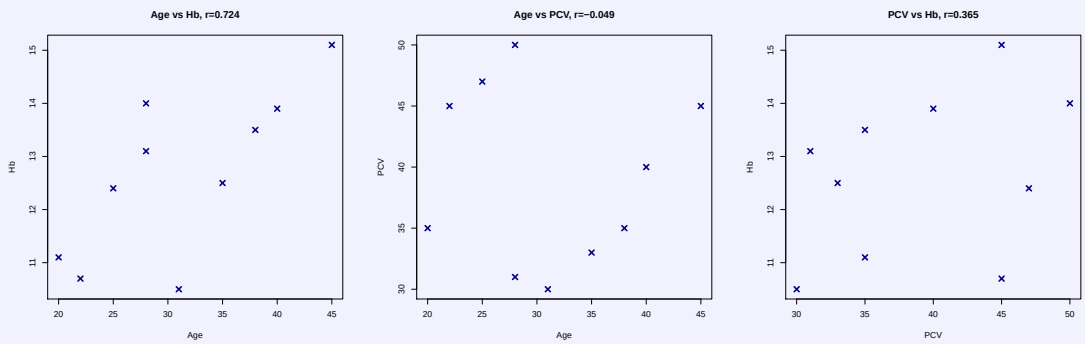

Figure 6: Scatterplots associated with Table 1

3 Ordinary Least Squares Regression

Any straight line has equation of the form $y = a + bx$. The line is determined by the values of a (the intercept) and b (the slope/gradient). Ordinary least squares regression aims to find the "best" values of a and b to "fit" a given set of points. There are various methods for determining this, but by far the most common method is to choose the "best" line according to the *principle of least squares* so as to minimize "the sum of the squares of the residuals", a residual being the difference between an observed value and the value predicted by the regression equation. There are various preconditions that the residuals should meet but we will not go into them here. In the past the values of a and b were calculated by hand using:

$$b = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (x - \bar{x})} \qquad a = \bar{y} - b\bar{x}$$

but nowadays they are invariably calculated using software packages such as `R`. If `R` is used to find the regression line associated with *Age* versus *Hb* from the example of Section 2.2.1 the following analysis is obtained:

```
> my_model = lm(formula = blood$Hb ~ blood$Age)
> my_model

Call:
lm(formula = blood$Hb ~ blood$Age)

Coefficients:
(Intercept)    blood$Age
     8.4229       0.1364
```

These are the values for a and b. We can get some more details by doing:

```
> summary(my_model)

Call:
lm(formula = blood$Hb ~ blood$Age)

Residuals:
     Min       1Q   Median       3Q      Max
-2.15271 -0.55083 -0.01627  0.55873  1.75663

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  8.42289    1.47670   5.704 0.000452 ***
blood$Age    0.13645    0.04594   2.970 0.017870 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 1.123 on 8 degrees of freedom
Multiple R-squared:  0.5244,    Adjusted R-squared:  0.465
F-statistic: 8.821 on 1 and 8 DF,  p-value: 0.01787
```

Note the following:

- The variable being predicted, here *Hb* is the *response* variable. The variable used for the prediction is the *independent* or *predictor* variable, or as a *covariate*.

- $Hb = 8.42 + 0.136Age$. The estimated *mean* value of Hb for a given value of Age is predicted to be that obtained by plugging Age into the equation, e.g. for $Age = 40$ we predict that, *on average*, $Hb = 8.423 + (0.13645 \times 40) = 13.881$

- `Residual standard error`$= 1.123$, this is the standard deviation of the residuals. More than this this implies that all the data is conditionally normally distributed with mean as predicted by the equation, and standard deviation 1.123. e.g. The Hb of 40 year olds is predicted to be normally distributed with mean 13.88 and standard deviation 1.123. The Hb of 21 year olds is predicted to be normally distributed with mean 11.29 and standard deviation 1.123.

- Note that from Table 1 we see that the minimum age is 20 and the maximum 45. We should be very cautious in making predictions outside of this age range ("extrapolating outside of the range of the observed data") as we cannot be sure that the pattern observed within this range will continue outside it. For example, if we plug $Age = 10$ in the regression equation we predict that the average value of Hb is 9.787, but it may well be the case that the relationship between Age and Hb differs in childhood to adulthood.

- The coefficient of Age is (estimated to be) 0.13645. This implies that for each unit increase in the value of age Hb is predicted to increase by, on average, 0.13645.

- The (estimated) constant "coefficient" (often referred to as the "intercept") is 8.423. Usually this has no interpretation other than if we were to extend the regression line until it intercepts the y axis, it would intercept at $y = 8.423$.

- `R-squared`$=0.5244$. This may be interpreted as saying that 52.4% of the variation in Hb is attributable to variation in age, the other 47.6% being attributable to other things not considered. This is also the square of the correlation coefficient; thus we could calculate that $r = \sqrt{0.524} = 0.724$. (As r is clearly positive, the correlation is the positive square root.)

Exercise

A commonly used measure of lung function is "FEV1" which stands for "forced expiratory volume in one second".

(i) Plot the data of Table 3 as a scatter diagram with height (cm) as x.

Table 3: Height and FEV1

Height (cm)	FEV1 (litres)	Height (cm)	FEV1 (litres)
164.0	3.24	176.0	3.75
167.0	3.24	177.0	4.05
170.4	3.19	177.4	3.60
171.2	3.42	180.7	4.80
171.3	3.20	181.0	3.96
172.0	3.60	183.1	4.78
172.0	3.78	183.6	4.56
174.0	4.32	183.7	4.68

(ii) Briefly explain the statement "the correlation between Height and FEV1 is 0.861".

(iii) The R output when FEV1 is regressed on height is:

```
> lung_model = lm(formula = FEV1 ~ Height)
> summary(lung_model)

Call:
lm(formula = FEV1 ~ Height)

Residuals:
     Min       1Q   Median       3Q      Max
-0.46304 -0.21927  0.00408  0.18612  0.54082

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -10.74759    2.31237  -4.648 0.000377 ***
Height        0.08349    0.01319   6.332 1.85e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3083 on 14 degrees of freedom
Multiple R-squared:  0.7412,    Adjusted R-squared:  0.7227
F-statistic: 40.09 on 1 and 14 DF,  p-value: 1.854e-05
```

What is the regression equation? Interpret the coefficients of -10.7476 and 0.083487.

(iv) Use the regression equation to predict the FEV1 of a student 174 cm tall. Briefly explain why your answer differs from the FEV1 of 4.32 litres observed for the student of height 174 cm in Table 3.

(v) What is the predicted *distribution* of FEV1 for students that are 174 centimetres tall?

(vi) Use the regression equation to predict the FEV1 of a student 205 cm tall. Comment on the reliability of this prediction.

Solution

(i)

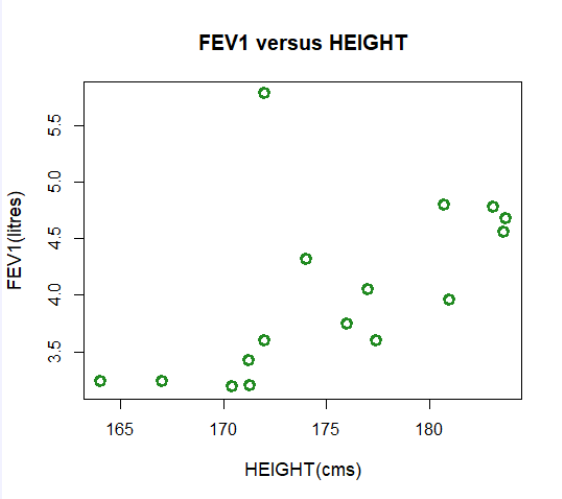

Figure 7: Question 1

(ii) Fairly strong positive relationship, as height increases so does FEV1, we may make reasonably precise predictions of the value of FEV1 for a given value of height.

(iii) The line of best fit (according to the "principle of least squares") is

$$FEV1 = -10.7 + 0.0835 Height$$

$a = -10.7476$ is the intercept, i.e. the value of FEV1 at which the line, continued back to height $= 0$, would cross the vertical axis.

$b = 0.083487$ is the gradient, i.e. the estimated mean increase in FEV1 for each 1cm increase in height.

(iv) If $x = 174$, $\hat{y} = -10.748 + (0.08349 \times 174) = 3.78$, i.e. 3.78 litres

This is the best prediction, based on the available data, of the mean FEV1 of students 174 cm tall. Any particular student of that height may have FEV1 above or below the average.

(v) Normally distributed with mean 3.78 and standard deviation 0.3083.

(vi) $x = 205 \Rightarrow y = 6.367$ i.e 6.37 litres.

Very unreliable, as we have no information for anybody over 183.7 cm tall.